Interacting with Data: Canada

This assignment uses the supplied 2021 Canadian population and health-inequality data. Parts 1, 2, and 3 are required for a B (3.2). Part 4 is an optional A-level extension and should be completed only after the required parts are correct.

General submission specifications

- Submit written answers and the public webpage link in one PDF.
- Submit one clearly labeled .py or .ipynb code file.
- Export each interactive visualization to the specified HTML file and preserve its interactive features.
- Check that files open correctly and use readable title-case labels, intentional colors, and honest scales.
- Code must run without errors using Folium and Plotly as specified.

> Use the supplied Excel and GeoJSON files. Do not replace them with newly downloaded versions. Treat every detailed specification as a checklist.

Part 1: Mapping Indigenous Population Composition

Remove the Canada-wide row. Correct Yukon to Yukon Territory in the population data, then match the population records to the GeoJSON feature names. Create Indigenous_Total and Total_Population. Calculate each recognized group as a percentage of the total population: Pct_First_Nations, Pct_Inuit, and Pct_Métis. Create formatted percent strings for the tooltip.

Task 1

Create a Folium choropleth with four switchable FeatureGroup layers in this order: Total Indigenous Population (raw count), First Nations (%), Inuit (%), and Métis (%).

- Display data for all 13 provinces and territories with no NaN areas.
- Center Canada at a useful default zoom and make sure only Canada and its subdivisions appear.
- Make sure the LayerControl behaves like radio buttons and show the layer control expanded.
- Use a color gradient that is suitable for the data range, and that highlights contrasts; The count layer must use raw counts; the other three layers must use percentages of the total population.
- Note that because we are using the feature groups, the color bar legend is going to disappear – we are not going to worry about this, as the relevant information is contained in the tooltip, and adding it back is overly complicated.
- Add the same hover tooltip to every feature group using a loop. Include location, total population, total Indigenous population, and the raw count plus percentage for First Nations, Inuit, and Métis.
- Use title-case reader-friendly aliases and preserve the accent in Métis.
- Save the interactive map as map.html.

Task 2

- Which regions have the largest total Indigenous populations? How does that differ from the regions with the largest Indigenous percentages?
- Where is each recognized group most geographically concentrated when measured as a percentage of the total population? Identify notable provinces or territories and describe the broad geographic pattern.

Part 2: Community Belonging and Mental Health Support

Use Canada Merge Data. Remove rows with 'Missing' or blank Domain values. Rename NonIndigenous [reference] as Non Indigenous and First Nations off reserve as First Nations. Use CR_BELONG for the percentage reporting strong or somewhat strong community belonging, and CR_HSC for the percentage reporting that all perceived mental-health needs were met. Remember that missing percentages should not automatically become zeros.

Task 1

Create an animated Plotly scatterplot with CR_BELONG on the x-axis, CR_HSC on the y-axis, Sex as the animation frame, and Domain mapped to both color and symbol.

- Use fixed, consistent x- and y-axis ranges that remain identical across animation frames.
- Use a neutral gray for Non-Indigenous and distinct colors and symbols for First Nations, Inuit, and Métis.
- Use Geography as the hover title. Show only the two plotted percentages formatted to two decimal places and the two relevant survey denominators (DENOM_BELONG and DENOM_HSC).
- Replace all coded column names with reader-friendly title-case labels.
- Use sufficient figure size and marker size, and reduce distracting non-data ink.
- Save scatter.html.

Task 2

- Using the plotted records what patterns do you see in community belonging alone? Compare populations and sexes.
- Using the plotted records what patterns do you see in mental-health needs met alone? Compare populations and sexes.
- Describe the relationship between the variables in each sex frame. Note outliers and the limited number of Inuit observations when interpreting the pattern.

Part 3: Integrated Interactive Webpage

Create one HTML page that presents map.html and scatter.html in order. Use a clear page title and a heading for each visualization, size both embedded frames so all controls remain visible, and preserve the interactive HTMLs. Publish the page and include its public link in the submitted written response.

Part 4: Optional A-Level Donut Subplots

Identify the three provinces or territories with the largest raw Indigenous_Total values. For those three locations, create one row of three Plotly donut subplots showing composition within the Indigenous population only: First Nations, Inuit, and Métis.

- Order locations from largest to smallest Indigenous_Total.
- Use the same group labels and colors used elsewhere; use spaces rather than underscores.
- Show percentages on the donuts, placing them inside where possible and outside when necessary.
- Place a centered annotation above each donut with the location name and Indigenous total count.
- Use one shared legend. Format each tooltip to show only the group name, raw count, and percentage within the Indigenous population.
- Save donuts.html.
- Add this to the webpage created in Part 3.
- In a few sentences, compare the Indigenous composition of the three locations.